

LOOPING STATEMENTS

Definition

Looping statements are used to execute a block of code repeatedly until a specified condition becomes false. They help reduce code repetition and make programs more efficient.

Introduction

- A loop repeats a set of instructions.
- It helps automate repetitive tasks.
- It reduces the amount of code.
- It improves program readability.
- It makes the program easier to maintain.
- Loops save time and reduce human error.

Types of Looping Statements

1. For Loop
2. While Loop
3. Do-While Loop

1. For Loop

Definition

The for loop is an entry-controlled loop where the condition is checked before the loop body executes. It is mainly used when the number of iterations is known.

Syntax

```
for(initialization; condition; increment/decrement){  
    // statements  
}
```

EXAMPLE

```
public class ForLoopExample {  
    public static void main(String[] args) {  
        for(int i = 1; i <= 5; i++) {  
            System.out.println(i);  
        }  
    }  
}
```

OUTPUT

```
1  
2  
3  
4  
5
```

2. While Loop

Definition

The while loop is an entry-controlled loop that executes as long as the condition is true. It is useful when the number of iterations is not known in advance.

Syntax

```
while(condition)  
{  
    // statements  
}
```

EXAMPLE

```
public class WhileLoopExample {  
    public static void main(String[] args) {  
        int i = 1;  
        while(i <= 5) {  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

OUTPUT

```
1  
2  
3  
4  
5
```

3. Do-While Loop

Definition

The do-while loop is an exit-controlled loop. The loop body executes first, and then the condition is checked. Therefore, the loop executes at least once.

Syntax

```
do  
{  
    // statements  
}  
while(condition);
```

EXAMPLE

```
public class DoWhileExample {  
    public static void main(String[] args) {  
        int i = 1;  
        do {  
            System.out.println(i);  
            i++;  
        } while(i <= 5);  
    }  
}
```

OUTPUT

```
1  
2  
3  
4  
5
```

Conclusion

Looping statements are an essential feature of Java programming. They make programs efficient by automating repetitive tasks. The for loop is ideal when the number of iterations is known, the while loop is suitable when the number of iterations is unknown, and the do-while loop is best when the code must execute at least once.